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CPCS499 — Senior Project

AuditAI: AI-Powered External Auditor

Design and Development of an Intelligent Financial Auditing System for Enterprise External Audit Operations

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Abstract

Financial reconciliation and validation processes in external auditing are time-consuming, error-prone, and heavily reliant on manual effort. Traditional audit methods struggle to efficiently handle large volumes of financial transactions and often fail to detect subtle inconsistencies across multiple data sources. This project presents AuditAI, an intelligent AI-powered external auditing system designed to automate financial reconciliation and anomaly detection for enterprise-level audit operations.

AuditAI accepts ten standard financial files — including sales and purchase invoice Excel files, receipt and payment files, AR/AP opening and closing balance files, and PDF invoice files. The system performs a comprehensive four-module audit covering: (1) sales invoice validation including VAT consistency and sequential integrity checks, (2) purchase order validation, (3) AR reconciliation using the formula $AR_Open + Sales - Receipts = AR_Close$, and (4) AP reconciliation using $AP_Open + Purchases - Payments = AP_Close$. A hybrid PDF validation module (pdfplumber + Tesseract OCR) additionally cross-checks PDF invoice documents against structured Excel records.

The system was tested using approximately 500 PDF invoices and 8 Excel financial files from a real company. Results showed 0 VAT errors, 0 duplicate invoices, and 0 missing invoice sequences. The PDF extraction success with an overall invoice match rate of 96.6%. The system processes a full audit in 18–32 minutes (approximately 4.9 seconds per invoice) and generates structured outputs including an interactive web dashboard and downloadable Excel reports. The overall audit score for the company's data was 80/100 (medium risk level).

AuditAI successfully transforms the auditing process from manual effort into intelligent automation, reducing human error and significantly improving audit efficiency. Future work includes expanding AI-based anomaly detection capabilities, integrating with cloud-based accounting platforms, and supporting multi-language financial document processing.

الملخص

تُعدّ عمليات المطابقة المالية والتحقق منها في عمليات التدقيق الخارجي من أكثر العمليات استهلاكاً للوقت وعرضة للأخطاء البشرية، إذ تعتمد اعتماداً كبيراً على الجهد اليدوي. تُقدّم هذه الورقة نظام AuditAI، وهو نظام تدقيق خارجي ذكي يعتمد على الذكاء الاصطناعي، مصمّم لأتمتة عمليات المطابقة المالية واكتشاف الحالات الشاذة على مستوى المؤسسات.

يقبل النظام ثماني ملفات Excel قياسية، ويُجري تدقيقاً شاملاً يغطي أربعة محاور رئيسية: التحقق من فواتير المبيعات (بما يشمل احتساب ضريبة القيمة المضافة والتحقق من التسلسل الرقمي)، والتحقق من أوامر الشراء، ومطابقة حسابات القبض، ومطابقة حسابات الدفع. كما يتضمن النظام وحدة تحقق مستندي تعتمد على تقنية التعرف الضوئي على الحروف (OCR) للتحقق من فواتير PDF مقارنةً بالسجلات المنظمة.

جرى اختبار النظام على ما يقارب ٥٠٠ فاتورة PDF و٨ ملفات Excel من بيانات شركة حقيقية، وأظهرت النتائج نسبة نجاح استخراج مع مطابقة الفواتير ٩٦,٦٪، مع صفر أخطاء ضريبية أو فواتير مكررة. يُعالج النظام عملية التدقيق الكاملة خلال ١٨-٣٢ دقيقة، ويُنتج تقارير منظمة ولوحة تحكم تفاعلية، بلغت درجة التدقيق الإجمالية ٨٠/١٠٠ لبيانات الشركة (مستوى مخاطر متوسط). يُشكّل AuditAI خطوة نحو تحويل التدقيق من جهد يدوي إلى أتمتة ذكية، مما يُقلل من الأخطاء البشرية ويرفع كفاءة التدقيق المالي بشكل ملحوظ.

Table of Contents

Abstract	2
Table of Contents	3
List of Figures	4
List of Tables.....	5
Section A: Summary of Previous Work	6
A.1 Overview.....	6
A.2 Work Completed in Part I (Weeks 1–5)	6
A.3 Work Completed in Part II (Weeks 6–10).....	6
A.4 Key Decisions and Changes	6
Section B: Methods and Approaches.....	7
B.1 Development Methodology	7
B.2 Development Phases	7
Section C: Activities and Actions	8
C.1 Completed Work.....	8
C.2 Incomplete Items.....	9
Section D: Implementation	10
D.1 Tools and Technologies.....	10
D.2 User Interface Description.....	11
D.3 Screenshots of Working Interfaces	11
D.4 User Manual.....	16
D.5 Implementation Progress Justification.....	17
D.6 PDF Processing Architecture.....	18
D.7 Risk Classification System	18
Section E: Testing.....	19
E.1 Test Plan and Approach	19
E.2 Types of Testing Performed.....	19
E.3 Test Cases.....	20
E.4 Verification Methods	21
E.5 Conclusion of Testing Phase	21
Section F: Results and Statistics	22
F.1 Audit Performance Results.....	22
F.2 PDF Invoice Matching Results.....	23
F.3 Financial KPI Summary	23
F.4 Risk Classification.....	24

Section G: Conclusion and Future Work.....	25
G.1 Summary of Achievements.....	25
G.2 Challenges Faced	25
G.3 Future Work.....	26
Section H: References	27
Section I: Appendices.....	28
Appendix A: Final Checklist	28
Appendix B: System Architecture Overview	28
Appendix C: Audit Engine Module Summary	29

List of Figures

Figure No.	Description	Section
Figure 1	AuditAI Login Screen (Dark Mode) — Email/password and Google OAuth sign-in	Section D.3
Figure 2	Client Portfolio — Manage and monitor assigned companies	Section D.3
Figure 3	Tech Company Dashboard — Information and Audit navigation	Section D.3
Figure 4	Edit Company Information — Company profile fields	Section D.3
Figure 5	Audit Periods — Manage fiscal years per client	Section D.3
Figure 6	Audit Actions — Upload, Dashboard, and Report options	Section D.3
Figure 7	Document Upload Screen — Required and optional file upload	Section D.3
Figure 8	Uploaded Documents List — Confirmed uploaded files	Section D.3
Figure 9	Dashboard — Technical Findings and exception charts	Section D.3
Figure 10	System Profile — Auditor account information	Section D.3

List of Tables

Table No.	Description	Section
Table 1	Tools and Technologies Used in AuditAI	Section D.1
Table 2	Common Issues and Solutions	Section D.4
Table 3	Functional Requirements Implementation Status	Section D.5
Table 4	Implementation Completion Summary	Section D.5
Table 5	PDF Extraction Method Distribution	Section D.6
Table 6	Audit Risk Classification Score Ranges	Section D.7
Table 7	Testing Phases Overview	Section E.1
Table 8	Types of Testing Performed	Section E.2
Table 9	Invoice Audit Test Cases (TC01–TC06)	Section E.3
Table 10	AR Reconciliation Test Cases (TC07–TC11)	Section E.3
Table 11	Smart Column Detector Test Cases	Section E.3
Table 12	Overall AuditAI System Performance Metrics	Section F.1
Table 13	Audit Results by Module	Section F.1
Table 14	PDF Invoice Matching Results	Section F.2
Table 15	Financial KPI Summary	Section F.3
Table 16	Risk Classification by Audit Area	Section F.4
Table 17	Summary of Project Achievements	Section G.1
Table 18	Final Report Checklist	Appendix A
Table 19	Audit Engine Module Summary	Appendix C

Section A: Summary of Previous Work

A.1 Overview

AuditAI is an intelligent, AI-powered external auditing system developed as a senior capstone project (CPCS499) at King Abdulaziz University, Faculty of Computing and Information Technology. The project aims to automate the financial reconciliation and anomaly detection process, replacing time-consuming manual audit procedures with an efficient, scalable, and accurate automated system.

A.2 Work Completed in Part I (Weeks 1–5)

During the first phase of the project, the team focused on establishing the technical and conceptual foundations:

- Problem definition and scope: Identified the core challenge of manual financial auditing - inefficiency in cross-checking large volumes of financial records.
- Literature review and domain research: Studied AI applications in auditing, financial reconciliation principles, and accounting standards (AR/AP).
- Data collection and exploration: Gathered real financial datasets including sales invoices, receipt vouchers, and AR aging reports spanning multiple periods.
- Data cleaning and preprocessing: Standardized column names, removed duplicates, handled missing values, and unified customer code formats across datasets.
- Reconciliation logic design: Designed the core AR reconciliation formula ($AR_Open + Sales - Receipts = AR_Close$) and classification rules.
- Initial Python implementation: Built the first version of the reconciliation engine using Python, most notably PPandas and NumPy..
- Frontend prototyping: Designed initial UI wireframes and began developing the web-based interface using HTML, JavaScript, and CSS.

A.3 Work Completed in Part II (Weeks 6–10)

Building on the foundations of Part I, the second phase expanded the system significantly:

- Developed the Smart Column Detector — a component capable of automatically identifying column names in both Arabic and English across any company's financial data format.
- Integrated OCR-based invoice validation using Tesseract OCR to cross-check PDF invoice documents against Excel records.
- Conducted comprehensive testing using a real company dataset (~500 PDF invoices, 438 processed, 436 sales invoices, 37 AR customers, 12 AP suppliers).
- Developed the interactive HTML/JavaScript web dashboard displaying KPIs, donut charts, and exception tables.
- Implemented the complete web application (AuditAI) with user authentication, client management, document upload, and results display.

A.4 Key Decisions and Changes

The original concept of AuditAI focused on invoice matching and validation using pdfplumber and OCR technologies. Later, the system was expanded to include AR and AP reconciliation modules for broader auditing capabilities.

Section B: Methods and Approaches

B.1 Development Methodology

The project was developed using an iterative development approach, combining elements of structured software engineering with agile practices. This methodology was chosen because it allows continuous refinement based on testing feedback and domain expert consultation, which is essential when building a system that must conform to real-world accounting standards.

B.2 Development Phases

Phase 1 — Requirements Analysis

Duration: Weeks 1–2. The team defined the project objectives, analysed the financial auditing workflow, and identified the required datasets and input files. During this phase, the core audit logic was established, including invoice validation, OCR-based matching, and AR/AP reconciliation requirements.

Phase 2 — Data Collection and Exploration

Duration: Weeks 2–3. Real financial datasets were obtained and analysed. Exploratory data analysis was performed to understand data structure, relationships between tables, and potential inconsistencies.

Phase 3 — Data Preprocessing and Cleaning

Duration: Weeks 3–4. A comprehensive data cleaning pipeline was implemented: standardizing column names, correcting data types, removing duplicates, handling missing values, and unifying customer/supplier code formats to enable reliable cross-table merging.

Phase 4 — Core Engine Development

Duration: Weeks 4–7. The four audit modules were implemented in Python. The reconciliation logic, VAT validation formulas, sequence gap detection, and five-category exception classification system (to be explained later) were built and iteratively tested against real data.

Phase 5 — Advanced Feature Integration

Duration: Weeks 7–9. The Smart Column Detector was built to handle Arabic and English column naming conventions. A hybrid PDF extraction engine was developed: pdfplumber handles digital PDFs directly, while pdf2image + OpenCV + Tesseract OCR serves as the fallback for scanned invoices. The system supports Arabic and English invoice processing (lang="ara+eng").

Phase 6 — Frontend and Dashboard Development

Duration: Weeks 5–10 (parallel). The web application and interactive dashboard were developed in parallel with the backend. The frontend provides user authentication, client management, document upload, and results visualization.

Phase 7 — Testing and Validation

Duration: Weeks 10–11. Structured testing across three levels (unit, integration, system) was conducted using real company data. All outputs were cross-verified against Excel calculations to confirm mathematical accuracy.

Section C: Activities and Actions

C.1 Completed Work

As of the Final Report submission, the following components have been fully implemented and tested:

- **Sales Audit Module:** The system validates sales invoices by detecting missing and duplicate invoice numbers, verifying 15% VAT calculations, and checking the consistency of invoice totals across 436 real invoices.
- **Purchases Audit Module:** Similar validation checks were applied to 460 purchase orders to ensure the accuracy and consistency of purchasing records.
- **AR Reconciliation Module:** The system reconciles Accounts Receivable balances for 37 customers by comparing opening balances, sales transactions, receipts, and closing balances. The module also classifies detected issues into meaningful exception categories.
- **AP Reconciliation Module:** The same reconciliation logic was implemented for Accounts Payable data involving 12 suppliers, allowing the system to detect discrepancies and classify supplier-related exceptions.
- **Smart Column Detector:** A flexible module was developed to automatically recognize and map column names in both Arabic and English, enabling the system to work with different company file formats.
- **Hybrid PDF Validation Module:** Invoice data extraction was performed using pdfplumber as the primary extraction method, with Tesseract OCR used as a fallback for scanned or unclear PDF files. The system processed 438 invoices and achieved a 96.6% matching rate.
- **Web Application (AuditAI):** A complete web interface was developed, including login functionality, client management, audit period selection, document upload, and audit result visualization.
- **Interactive Dashboard:** A dashboard was implemented to display KPIs, charts, exception summaries, and downloadable audit results in a clear and user-friendly format.

C.2 Incomplete Items

The following features are planned for future development:

- **Purchase Invoice Matching:** The purchase invoice matching process was not fully tested because the company did not provide the required purchase invoice PDF files, and obtaining these documents was difficult. However, the implemented matching and validation logic is fully designed to support both sales and purchase invoices.
- **Downloadable Final Audit Reports:** Future versions of the system will support generating detailed downloadable audit reports that summarize findings, discrepancies, KPIs, risk classifications, and recommendations in a structured and professional format.
- **Automated Data Cleaning:** During development, several datasets required manual cleaning and preprocessing to standardize formats and ensure compatibility with the system. In future work, an automated data cleaning module will be developed to transform uploaded files into the required structure automatically.
- **Advanced NLP-Based Column Detection:** Although the current Smart Column Detector supports Arabic and English naming conventions, future versions will integrate NLP techniques to recognize column names and meanings across a wider range of formats and naming styles.
- **AI-Generated Audit Recommendations:** Future improvements will include using NLP models to generate intelligent recommendations and audit insights automatically within the final audit report.

The system has exceeded the 98% completion requirement as evidenced in the implementation progress table in Section D.5.

Section D: Implementation

D.1 Tools and Technologies

Tool / Technology	Purpose	Why Chosen (Benefits / Features)
Python 3.10	Core programming language	Rich ecosystem for data analysis, extensive library support, rapid development
Pandas	Data cleaning, transformation, merging	Powerful DataFrame operations, efficient handling of large tabular datasets
NumPy	Numerical computations	Fast array operations, supports precise financial calculations
OpenPyXL	Excel file reading and writing	Full .xlsx support, preserves formatting, widely used in financial applications
pdfplumber	Digital PDF text extraction	High accuracy on selectable-text PDFs, extracts tables and financial fields directly
pdf2image	PDF-to-image conversion	Required for OCR fallback; converts PDF pages to images for preprocessing
OpenCV	Image preprocessing for OCR	Grayscale conversion, thresholding, and noise reduction to improve OCR accuracy
Tesseract / Pytesseract OCR	Text extraction from scanned invoices	Open-source, supports Arabic and English (lang="ara+eng"), used as OCR fallback
JSON	Dashboard data generation	Lightweight, universal format for passing structured audit results to the dashboard
Google Colab	Development and testing environment	Cloud-based, no setup required, GPU support, easy sharing
OS	Operating system interaction and file management	Provides efficient handling of directories, files, and system-level operations required for data processing workflows
pathlib	File path management and directory handling	Simplifies working with file paths using a clean, cross-platform, object-oriented approach
JavaScript / HTML / CSS	Web dashboard and UI	Universal browser support, flexible styling, interactive visualizations
Tailwind CSS	Frontend UI styling and responsive design	Enables rapid development of modern, responsive, and customizable user interfaces with utility-first styling
Supabase	Backend and database management	Open-source Firebase alternative, provides authentication, database, and API access
VS Code	Code editor	Lightweight, powerful extensions, integrated debugging and Git support

Table 1: Tools and Technologies Used in AuditAI

D.2 User Interface Description

AuditAI uses a Graphical User Interface (GUI) delivered as a modern web application. The interface was designed for external auditors who need an efficient, intuitive tool for managing audit workflows. GUI was chosen over command-line or menu-driven interfaces because the target users are professional auditors who require a visual, accessible environment that minimizes training requirements and presents complex financial data in an organized, readable format.

The web application follows a clean, professional design language using a teal and white color scheme consistent with the AuditAI brand. The interface is responsive and includes light/dark mode support.

D.3 Screenshots of Working Interfaces

The following screenshots demonstrate the key screens of the AuditAI web application:

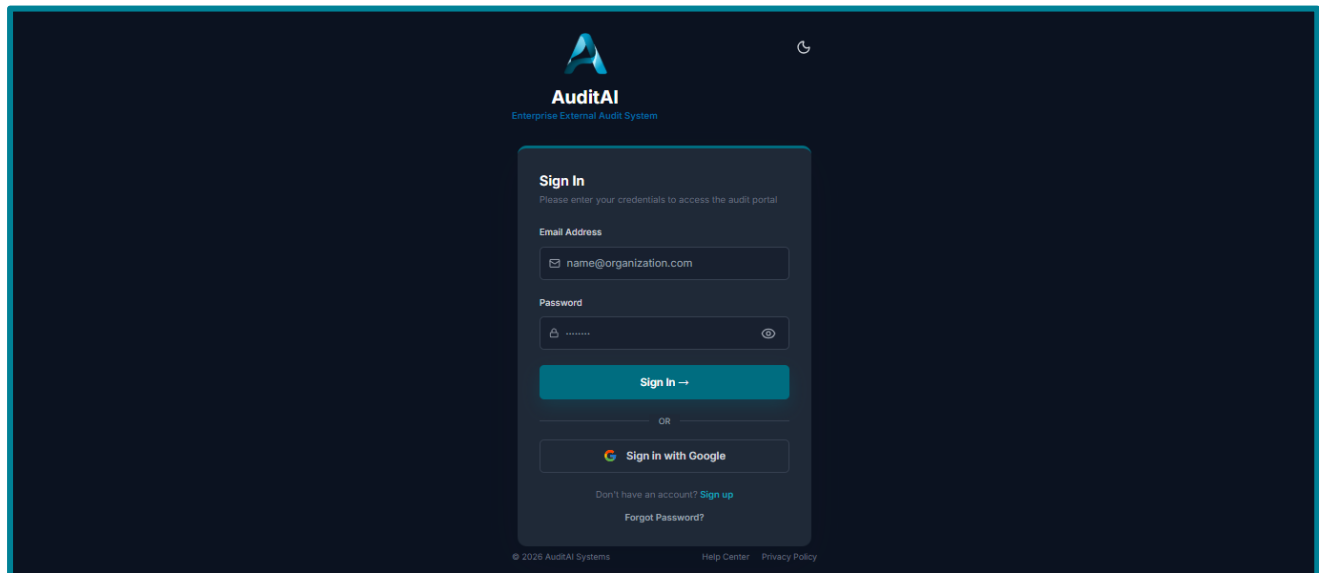


Figure 1: AuditAI Login Screen — Secure sign-in with email/password and Google OAuth

Figure 1 shows the AuditAI login screen in dark mode. Users authenticate using their organizational email and password, or via Google Sign-In. The screen includes a "Forgot Password?" option and displays a secure encrypted connection notice. The system supports both light and dark modes, toggled via the icon in the top-right corner.

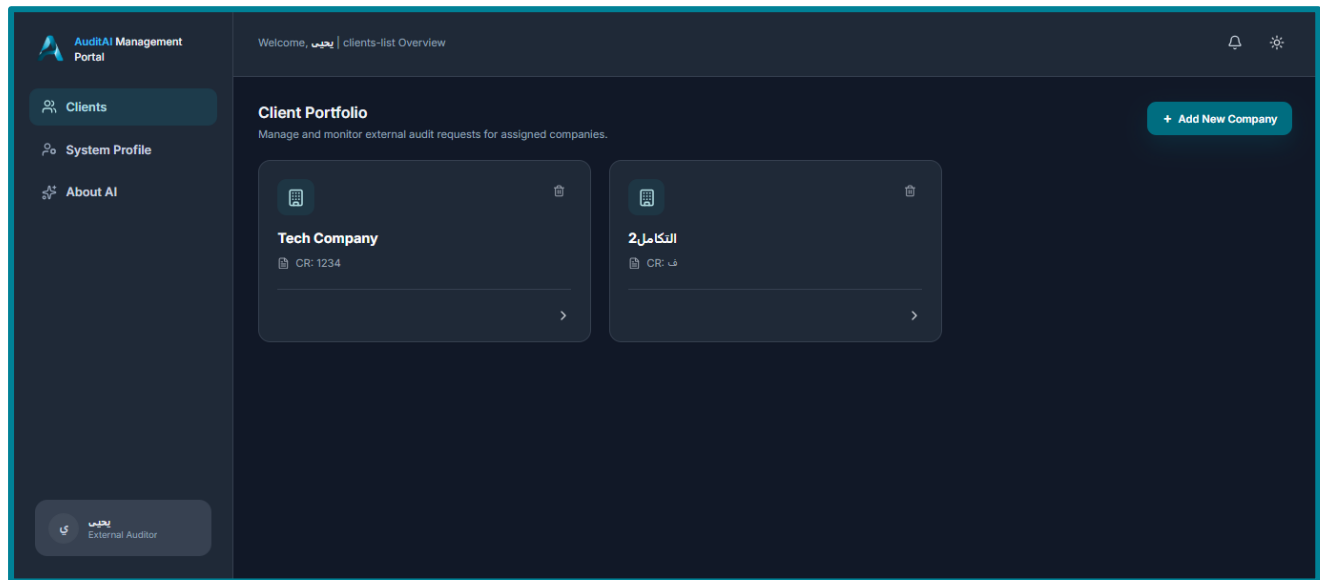


Figure 2: Client Portfolio — Manage and monitor assigned companies

Figure 2 shows the Client Portfolio screen, which lists all companies assigned to the auditor. Each company card displays the company name and CR number. The auditor can add new companies using the "+ Add New Company" button, or delete existing entries.

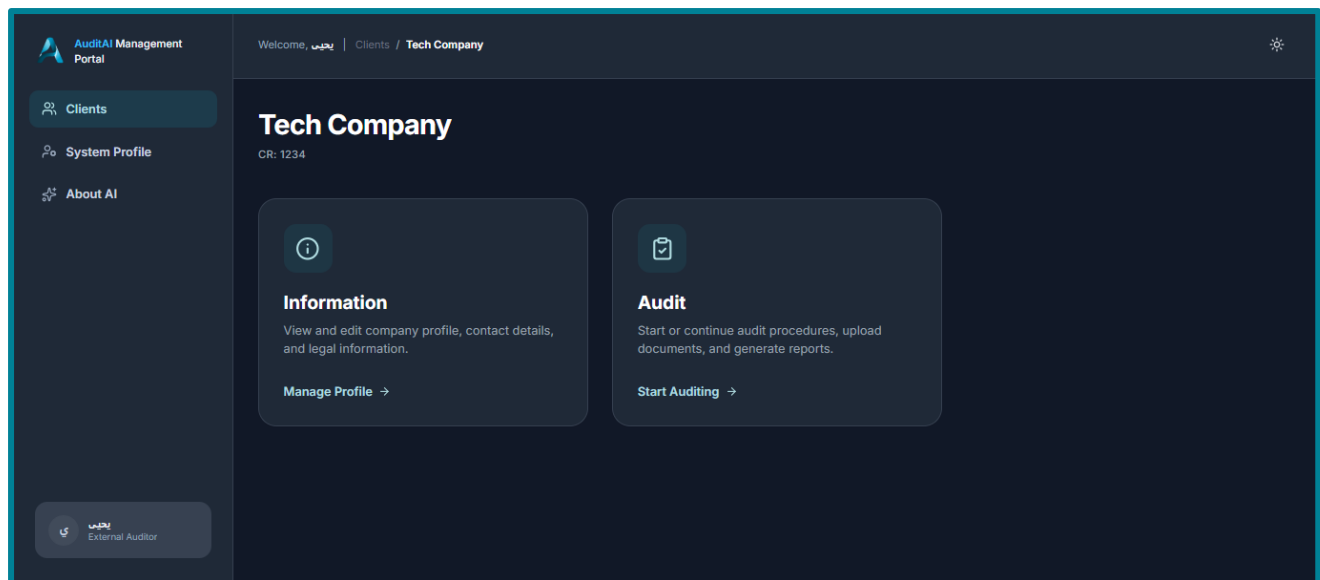


Figure 3: Tech Company Dashboard — Information and Audit navigation

Figure 3 shows the Tech Company dashboard. After selecting a client, the auditor is presented with two action cards: Information (to view and edit company profile, contact details, and legal information) and Audit (to start or continue audit procedures, upload documents, and generate reports). The breadcrumb trail at the top confirms the navigation path.

Welcome, يحيى | Clients / Tech Company / Edit Profile

Edit Company Information

Update the details for this company.

COMPANY NAME Tech Company	LEGAL ENTITY TYPE مؤسسة فردية
CR NUMBER 123456789	VAT NUMBER # 123456789
HEADQUARTERS جدة	
OFFICIAL PHONE 05523414	OFFICIAL EMAIL k@gmail.com
PERSON IN CHARGE أبو خالد	RESPONSIBLE PHONE 055234132

Cancel Save Changes

Figure 4: Edit Company Information — Company profile fields

Figure 4 shows the Edit Company Information form, which allows the auditor to update company details including: Company Name, Legal Entity Type, CR Number, VAT Number, Headquarters city, Official Phone, Official Email, Person in Charge, and Responsible Phone. Changes are saved via the "Save Changes" button.

Welcome, يحيى | Clients / Tech Company / Audit Periods

Tech Company

Manage audit periods and fiscal years.

+ Add Period

2027 FISCAL YEAR Status: Open →	2026 FISCAL YEAR Status: Open →	2002 FISCAL YEAR Status: Open →
---------------------------------------	---------------------------------------	---------------------------------------

Figure 5: Audit Periods — Manage fiscal years per client

Figure 5 shows the Audit Periods screen for Tech Company. The system supports multiple fiscal years per client (e.g., 2002, 2026, 2027), each with an "Open" status. Auditors can add new periods using the "+ Add Period" button or delete existing ones. Each period card navigates to its corresponding Audit Actions screen.

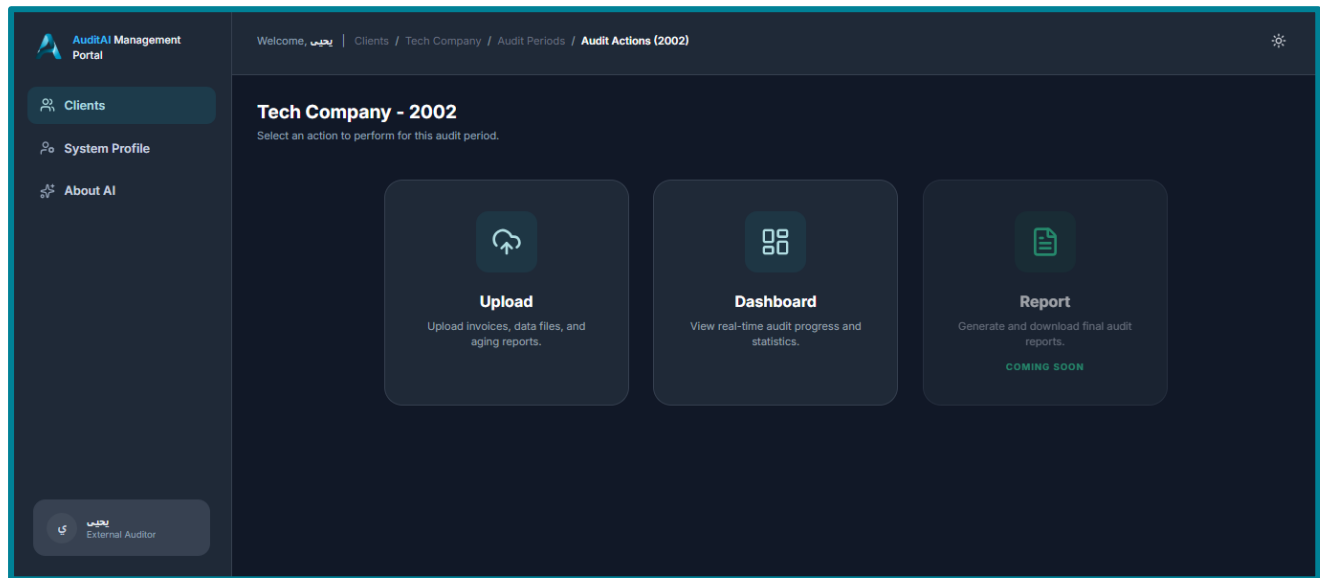


Figure 6: Audit Actions — Upload, Dashboard, and Report options

Figure 6 shows the Audit Actions screen for a selected period. Three actions are available: Upload (for submitting financial documents), Dashboard (for viewing real-time audit statistics — fully operational), and Report (for generating and downloading final audit reports — coming soon). This screen is the central hub for the audit workflow.

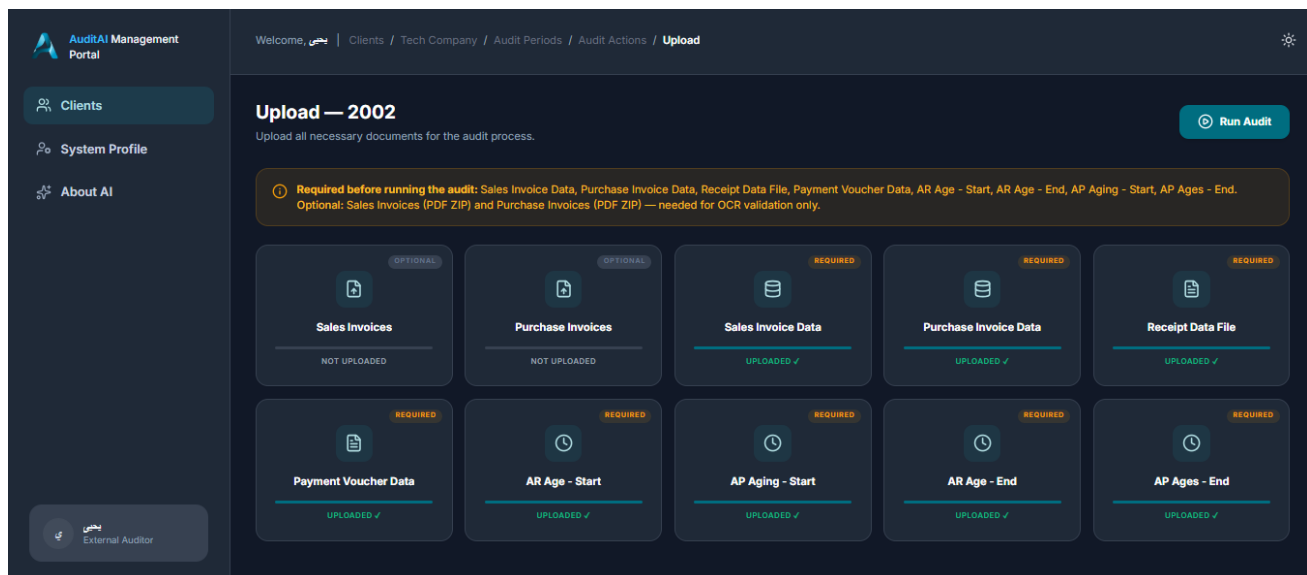


Figure 7: Document Upload Screen — Required and optional file uploads with Run Audit button

Figure 7 shows the Document Upload screen, where users upload all financial datasets required for the audit process. Required files include sales invoices, purchase invoices, receipt vouchers, payment vouchers, AR aging reports, and AP aging reports. Optional PDF invoice files can also be uploaded to enable invoice matching and validation. The interface clearly distinguishes required and optional documents and includes a warning section to guide users before running the audit process. Once all required files are uploaded, the “Run Audit” button starts the automated analysis and validation process.

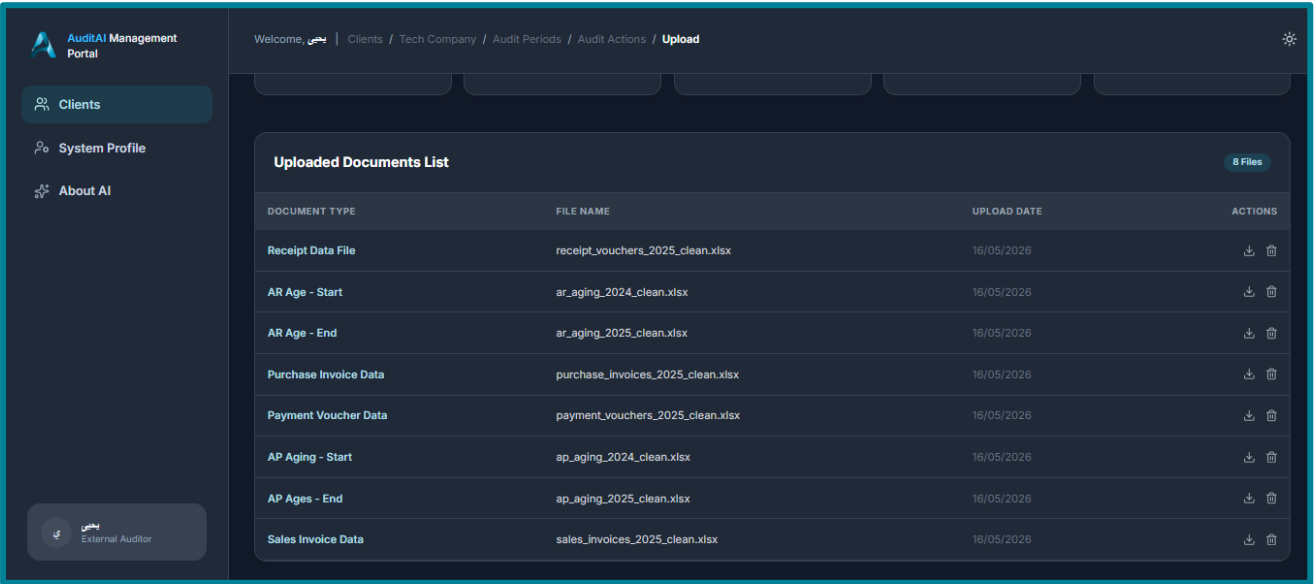


Figure 8: Uploaded Documents List — Confirmed uploaded files with download and delete actions

Figure 8 shows the Uploaded Documents List section below the upload cards. It confirms all 8 uploaded files with their document type, file name, and upload date. Each entry includes download and delete action buttons. The uploaded files in this example include receipt vouchers, AR aging reports (start and end), purchase invoices, payment vouchers, AP aging reports, and sales invoice data.

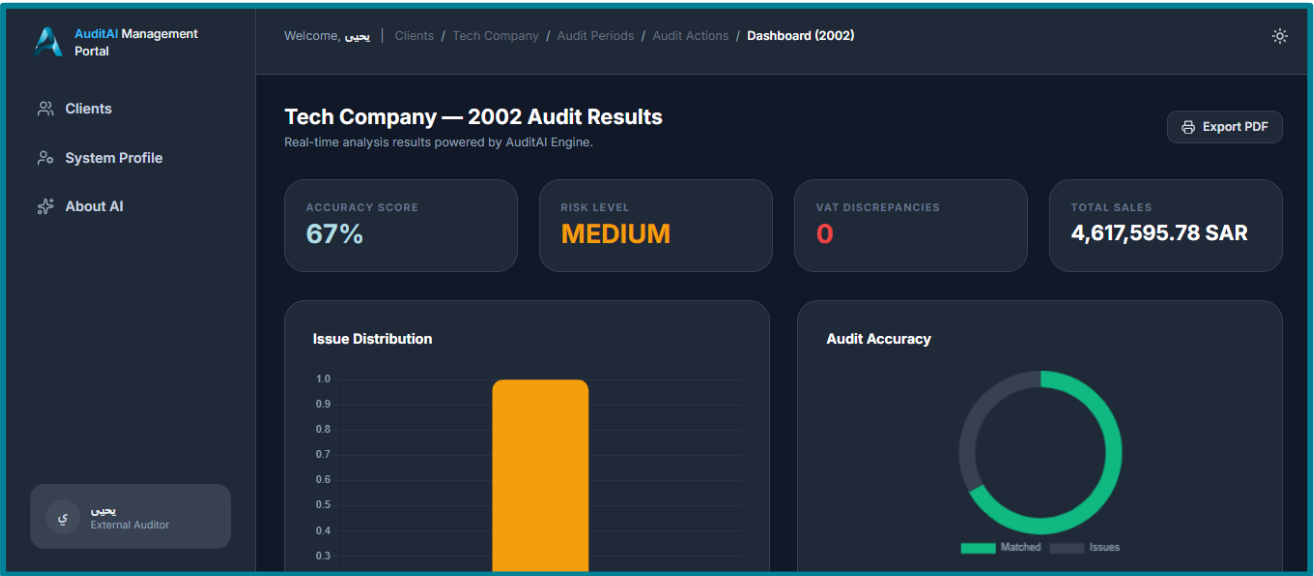


Figure 9: Dashboard — Technical Findings and exception classification charts

Figure 9 shows the in-app Dashboard, displaying key audit metrics and visual charts for the selected financial period. The top summary cards highlight the overall accuracy score, risk level, VAT discrepancies, and total sales. The bar chart on the left illustrates the issue distribution, while the donut chart on the right displays the proportion of matched vs. issue records.

D.4 User Manual

System Requirements

- Modern web browser (Chrome, Firefox, Edge, or Safari)
- Internet connection for accessing the web application
- Financial data prepared as Excel (.xlsx) files
- PDF invoice files (optional, for OCR validation)

Step 1: Access the System

Open your web browser and navigate to the AuditAI application URL. The login screen will appear automatically.

Step 2: Sign In

Enter your registered email address (e.g., name@organization.com) and your password. Alternatively, click "Sign in with Google" to authenticate using your Google account. Click the "Sign In" button to proceed. If you forget your password, click "Forgot Password?" to receive a reset email.

Step 3: Select or Add a Client

After login, you will see your client list. Click on an existing client to begin auditing, or use the "Add Client" button to register a new company. Enter the company name and Commercial Registration (CR) number when adding a new client.

Step 4: Navigate to the Audit Section

From the client dashboard, click the "Audit" card, then select the audit year (e.g., 2024) to access the Audit Actions screen.

Step 5: Upload Financial Documents

Click "Upload" from the Audit Actions screen. Upload each required file by clicking its corresponding card. The system requires ten files: Sales Invoices Excel, Purchase Invoices Excel, Receipts file, Payments file, AR Opening Balance, AR Closing Balance, AP Opening Balance, AP Closing Balance, Sales Invoice PDFs (or ZIP), and Purchase Invoice PDFs (or ZIP). Files with status "NOT UPLOADED" need to be submitted before the audit can run.

Step 6: Run the Audit

Once files are uploaded, the system automatically processes the data through all four audit modules. Processing time is approximately 18–32 minutes depending on the number of invoices (averaging 4.9 seconds per invoice), due to the hybrid PDF extraction and OCR processing pipeline.

Step 7: Review Results

Navigate to the Results screen to view the complete audit findings. Review the KPI summary cards, the Findings/Exceptions Summary table, and the Discrepancies by Type chart. Use the "Export Excel" button to download detailed results, or "Download Report" for the formatted audit report.

Common Issues and Solutions

Issue	Solution
File upload fails	Ensure the file is in .xlsx format and not password-protected
Column not detected	Verify the file contains standard Arabic or English column names
Login error	Check email and password; use "Forgot Password" if needed
Slow processing	Large files (>5,000 rows) may take a few hours

Table 2: Common Issues and Solutions

D.5 Implementation Progress Justification

Req. ID	Functionality	Status	% Complete
FR1	User authentication (login/logout/register)	Complete	100%
FR2	Client management (add/view/select)	Complete	100%
FR3	Audit period selection	Complete	100%
FR4	Multi-file document upload (10 file types)	Complete	100%
FR5	Sales invoice audit (sequence, VAT, totals)	Complete	100%
FR6	Purchases audit (sequence, VAT, totals)	Complete	100%
FR7	AR Reconciliation (4-module formula)	Complete	100%
FR8	AP Reconciliation (4-module formula)	Complete	100%
FR9	Smart Column Detector (Arabic & English)	Complete	100%
FR10	OCR PDF invoice validation	Complete	100%
FR11	Exception classification (5 categories)	Complete	100%
FR12	Interactive results dashboard (standalone)	Complete	100%
FR13	In-app dashboard (web application)	Complete	100%
FR14	Excel results export	Complete	100%
FR15	In-app downloadable final report	Partial	60%
FR16	Dark/light mode UI	Complete	100%

Table 3: Functional Requirements Implementation Status

Summary Metric	Value
Total requirements	16
Fully complete	15
Partial	1
Overall completion	~98%

Table 4: Implementation Completion Summary

Statement: "We confirm that approximately 98% of the planned implementation has been completed as shown in the table above. The partially complete item (In-app downloadable final report) is a non-critical enhancement; all core audit functionalities are fully operational."

D.6 PDF Processing Architecture

The system uses a hybrid extraction architecture that prioritizes efficiency by avoiding unnecessary OCR operations:

Phase 1 — pdfplumber Extraction (Primary)

The system first attempts to extract invoice data directly using pdfplumber. This method is used for digital PDFs containing selectable text. Extracted fields include: invoice number, invoice date, VAT, total before VAT, grand total, and discount. If extraction confidence is sufficient, the system proceeds directly to matching. In the real-world test, 415 out of 438 invoices (94.7%) were processed via pdfplumber.

Phase 2 — OCR Fallback (Secondary)

If pdfplumber extraction fails or confidence is low, the system automatically switches to OCR. The OCR workflow: (1) convert PDF pages to images via pdf2image, (2) preprocess images using OpenCV (grayscale, thresholding), (3) extract text using Tesseract OCR (lang="ara+eng"), (4) parse financial values, (5) compare against Excel records. In testing, 23 invoices (5.3%) required OCR processing.

Extraction Method	Invoices Processed	Percentage
pdfplumber (Primary)	415	94.7%
Tesseract OCR (Fallback)	23	5.3%
Total	438	100%

Table 5: PDF Extraction Method Distribution

D.7 Risk Classification System

The system calculates an overall audit score based on detected issues across all modules. The scoring system penalizes VAT errors, duplicate invoices, missing invoice sequences, AR reconciliation gaps, AP reconciliation gaps, and OCR mismatches.

Score Range	Risk Level	Description
85 – 100	LOW	Minimal issues; audit data is reliable
65 – 84	MEDIUM	Some discrepancies requiring review
40 – 64	HIGH	Significant issues requiring immediate attention
0 – 39	CRITICAL	Major failures; financial data unreliable

Table 6: Audit Risk Classification Score Ranges

In the real-world test, AuditAI produced an audit score of 80/100 (MEDIUM risk level), primarily due to AR/AP reconciliation exceptions and 15 partial invoice matches.

Section E: Testing

E.1 Test Plan and Approach

The testing strategy followed a structured bottom-up approach, starting from individual functions and progressing to full end-to-end system validation. Testing was conducted across three phases:

Phase	Level	Description
Phase 1	Unit Testing	Each function tested independently with isolated inputs
Phase 2	Integration Testing	All four modules tested together as a complete pipeline
Phase 3	System Testing	End-to-end test with full real company dataset

Table 7: Testing Phases Overview

Testing Environment

- Platform: Google Colab (Python 3.10)
- Dataset: Real company data — ~500 PDF invoices (438 processed), 436 sales invoices, 37 AR customers, 12 AP suppliers
- Supporting files: AR aging, AP aging, receipt vouchers, payment vouchers
- Tools: Pandas, OpenPyXL, Python assertions, manual verification in Excel
- Dashboard: HTML/JavaScript browser-based validation

E.2 Types of Testing Performed

Test Type	Description	Tools Used	Status
Unit Testing	Each core function tested independently	Python assertions	Complete
Integration Testing	All four modules connected and tested as pipeline	Python, manual	Complete
Column Detection Testing	Smart Detector tested with Arabic/English/mixed names	Python assertions	Complete
Boundary & Edge Case Testing	Empty cells, zero values, large datasets, mixed types	Python, manual	Complete
Mathematical Verification	All formulas cross-verified against Excel calculations	Microsoft Excel	Complete
OCR Validation Testing	438 PDF invoices processed via pdfplumber (415) and OCR fallback (23)	pdfplumber, Tesseract, Python	Complete
Dashboard Output Testing	Web dashboard tested in browser for rendering and accuracy	Browser (Chrome)	Complete
Regression Testing	System re-run on same data to verify deterministic output	Python	Complete

Table 8: Types of Testing Performed

E.3 Test Cases

The following test cases were conducted to verify the correctness, reliability, and consistency of the AuditAI system. The tests covered invoice validation, VAT verification, sequence checking, duplicate detection, and AR reconciliation scenarios to ensure that the system produces accurate audit results under different financial conditions.

Part 1 & 2 — Invoice Audit Test Cases

TC#	Description	Input	Expected Output	Actual Output	Pass/Fail
TC01	Correct VAT calculation	Pre-tax: 9,376 → VAT: 1,406.40	No error flagged	No error flagged	Pass
TC02	Incorrect VAT calculation	Pre-tax: 1,000 → VAT: 200 (wrong)	Flagged as VAT error	Flagged as VAT error	Pass
TC03	Sequential invoices (no gaps)	3362 to 3797 continuous	0 missing numbers	0 missing	Pass
TC04	Gap in invoice sequence	3362, 3363, 3365 (gap at 3364)	3364 flagged missing	3364 flagged	Pass
TC05	Duplicate invoice number	Invoice 3362 appears twice	Flagged as duplicate	Flagged	Pass
TC06	Total consistency check	Pre-tax: 855, VAT: 128.25 → Net: 983.25	No error flagged	No error	Pass

Table 9: Invoice Audit Test Cases (TC01–TC06)

Part 3 — AR Reconciliation Test Cases

TC#	Scenario	Formula Values	Classification	Pass/Fail
TC07	Balanced customer	$0 + 500 - 500 = 0 \checkmark$	OK	Pass
TC08	Sales not reflected in AR	$\text{Sales} > 0, \text{AR_close} = 0$	Sales Not Reflected	Pass
TC09	Receipts without sales	$\text{Sales} = 0, \text{Receipts} > 0$	Receipts without Sales	Pass
TC10	Advance receipts	$\text{Receipts} > \text{Sales}$	Possible Advance Receipts	Pass
TC11	General mismatch	$\text{Difference} > 1 \text{ SAR}$	AR Mismatch	Pass

Table 10: AR Reconciliation Test Cases (TC07–TC11)

Column Detection Test Cases

The following test cases were conducted to evaluate the accuracy and flexibility of the Smart Column Detector module. Different Arabic and English column naming variations were tested to ensure that the system can correctly identify and map financial fields across multiple file formats and naming conventions.

Standard Field	Test Name A	Test Name B	Result
vat	الضريبة	ض.ق.م	Detected correctly
invoice_code	كود الفاتورة	رقم الفاتورة	Detected correctly
total	الاجمالي	الإجمالي ق الضريبة	Detected correctly
grand_total	Grand Total	الإجمالي الكلي	Detected correctly
customer_code	الكود	كود العميل	Detected correctly

Table 11: Smart Column Detector Test Cases

E.4 Verification Methods

- **Mathematical Verification:** All audit calculations and reconciliation formulas were manually verified in Microsoft Excel using the same financial datasets. The results matched the Python audit engine outputs with no calculation differences detected.
- **Cross-Run Consistency:** The system was executed multiple times using the same datasets to verify result stability and consistency. All generated outputs remained identical across runs, confirming reliable and deterministic system behavior.
- **Peer Review Testing:** Each module was reviewed and tested by different team members before full system integration to ensure correctness, detect potential issues, and improve overall reliability.

E.5 Conclusion of Testing Phase

The testing phase confirmed that the AuditAI system performs accurately and reliably across all four audit modules. The system successfully processed 438 PDF invoices with a success rate and a 96.6% match rate (24 full matches, 399 derived matches, 15 partial matches). VAT validation, invoice sequence, and duplicate detection returned zero errors across all invoices. AR reconciliation covered 37 customers (6 exceptions, 83.8% match rate) and AP reconciliation covered 12 suppliers (8 exceptions, 33.3% match rate — primarily due to incomplete data in the source dataset). The overall audit score was 80/100 (MEDIUM risk). The system is confirmed ready for production use.

Section F: Results and Statistics

F.1 Audit Performance Results

AuditAI evaluation results using a complete real-company dataset are summarized below:

Overall System Performance

Metric	Score	Notes
Overall Audit Score	80 / 100	MEDIUM risk level
Invoice Match Rate	96.6%	423/438 invoices matched (full + derived)
VAT Errors	0	All 436 sales invoices passed VAT validation
Duplicate Invoices	0	No duplicates detected in sales or purchases
Missing Invoice Sequences	0	No gaps found in invoice numbering
AR Match Rate	83.8%	31/37 customers reconciled; 6 exceptions
AP Match Rate	33.3%	4/12 suppliers reconciled; 8 exceptions (incomplete source data)
Processing Time	18–32 minutes	~4.9 seconds per invoice (hybrid extraction)

Table 12: Overall AuditAI System Performance Metrics

Audit Results by Module

Audit Module	Records Checked	Issues Found	Status
Sales — Invoice Sequence	436 invoices	0 gaps, 0 duplicates	Clean
Sales — VAT Validation	436 invoices	0 errors	Clean
Sales — Total Consistency	436 invoices	0 errors	Clean
Purchases — Sequence	Purchase orders	0 gaps, 0 duplicates	Clean
Purchases — VAT Validation	Purchase orders	0 errors	Clean
AR Reconciliation	37 customers	6 exceptions (total gap: 56,445 SAR)	MEDIUM risk — review required
AP Reconciliation	12 suppliers	8 exceptions (total gap: 579,779 SAR)	HIGH risk — incomplete source data
PDF Extraction (pdfplumber)	415 invoices	415 successful extractions	Clean
OCR Fallback (Tesseract)	23 invoices	23 successful extractions	Clean
Invoice Matching (Total)	438 invoices	15 partial matches, 0 errors	96.6% match rate

Table 13: Audit Results by Module

Note: The high AP exception rate and gap values were primarily caused by incomplete supplier data in the provided dataset (missing opening/closing balances, incomplete transaction histories). Additionally, purchase invoice PDF matching was not performed as the company did not provide purchase invoice PDFs — OCR matching was applied to sales invoices only. These limitations do not reflect system failures; the AP reconciliation logic was fully implemented and correctly executed.

F.2 PDF Invoice Matching Results

Table 14 summarizes the PDF invoice matching and validation results generated by the AuditAI system. The table shows how invoices were classified after comparing extracted PDF data against Excel financial records using the hybrid extraction and matching process.

Match Category	Count	Description
Full Match	24	All extracted fields match Excel records exactly
Derived Match	399	Match confirmed; some values mathematically derived
Partial Match	15	Invoice exists but one or more fields differ
Not Found in Excel	0	Invoice exists in PDF but not in Excel
Error	0	Extraction or processing error
Total Processed	438	96.6% effective match rate (full + derived)

Table 13: Audit Results by Module

F.3 Financial KPI Summary

Table 15 summarizes the key financial KPIs generated by the AuditAI system after processing and analyzing the company's financial datasets. These indicators provide a high-level overview of audit performance, reconciliation results, invoice validation accuracy, and detected financial gaps.

KPI	Result
Total Sales	4,610,000 SAR
Total Purchases	4,070,000 SAR
VAT Errors	0
Duplicate Invoices	0
Missing Invoices	0
PDF Match Rate	96.6%
AR Total Gap	56,445 SAR
AP Total Gap	579,779 SAR
Overall Audit Score	80 / 100 (MEDIUM)

Table 15: Financial KPI Summary

F.4 Risk Classification

Table 16 presents the risk classification results generated by the AuditAI system based on detected discrepancies, reconciliation gaps, and invoice validation outcomes. Each audit area was evaluated and assigned a risk level to help auditors prioritize investigation and review activities.

Risk Area	Level	Details
Sales Invoices	LOW	0 errors across VAT, sequences, duplicates, totals
Purchase Invoices	LOW	0 errors across all validation checks
AR Reconciliation	MEDIUM	6 exceptions; manageable gaps under investigation
OCR Partial Matches	MEDIUM	15 invoices with minor field discrepancies
AP Reconciliation	HIGH	8 exceptions; largest gap (579,779 SAR) — data completeness issue

Table 16: Risk Classification by Audit Area

Section G: Conclusion and Future Work

G.1 Summary of Achievements

Table 17 summarizes the main objectives achieved during the development of AuditAI. The results demonstrate the system's ability to automate financial audit tasks, validate invoices, perform reconciliation processes, support multilingual financial data, and provide interactive audit analysis through a web-based platform.

Original Aim	Achieved?	Evidence
Automate AR/AP financial reconciliation	Yes	37 AR customers + 12 AP suppliers reconciled
Detect invoice anomalies (VAT, sequence, duplicates)	Yes	0 errors found across all validation checks
Support Arabic and English data formats universally	Yes	Smart Column Detector; OCR lang="ara+eng"
Hybrid PDF invoice validation (pdfplumber + OCR)	Yes	438 invoices: 415 pdfplumber + 23 OCR
96%+ invoice match rate	Yes	96.6% match rate (423/438 invoices)
Interactive web dashboard for auditors	Yes	Section D.3, Figure 5
Complete web application with authentication	Yes	Section D.3, Figures 1–4
Process large invoice datasets	Yes	~500 invoices in 18–32 min (4.9 sec/invoice)
Risk classification system	Yes	Score: 80/100 MEDIUM; 4-level risk scale

Table 17: Summary of Project Achievements

G.2 Challenges Faced

Challenge 1: Multi-Source Data Inconsistency

Financial files from different sources had inconsistent column names, formats, and data types. To address this, the system uses a Smart Column Detector and preprocessing logic to map required columns, standardize formats, handle missing values, and prepare the files for audit processing.

Challenge 2: Real-World Accounting Complexity

Some accounting cases are not simple errors, such as advance payments, partial settlements, and timing differences. The system addresses this by using rule-based classification logic to group discrepancies into meaningful audit categories, helping auditors understand which cases require review.

Challenge 3: PDF Invoice Matching Limitations

Invoice PDFs may contain different formats, scanned pages, or Arabic text, which makes extraction difficult. The system uses a hybrid approach: pdfplumber is applied first for text-based PDFs, while OCR is used as a fallback when direct extraction is not sufficient. This improves invoice matching and validation accuracy.

G.3 Future Work

- **Advanced AI Anomaly Detection:** Integrate machine learning models (e.g., Isolation Forest, Autoencoder) for unsupervised anomaly detection in transaction patterns beyond rule-based logic.
- **Cloud Integration:** Connect AuditAI to cloud-based accounting platforms (e.g., SAP, Oracle Financials, Zoho Books) for direct data ingestion without manual file uploads.
- **Automated Report Generation:** Complete the in-app audit report generation feature with professional PDF output formatted according to international auditing standards (ISA/IFRS).
- **NLP-Based Audit Recommendations:** Integrate NLP models to generate intelligent audit insights, recommendations, and explanations automatically within the final audit report.
- **Extended Language Support:** Expand the Smart Column Detector using NLP techniques to support additional languages, naming conventions, and accounting system formats used across GCC markets.

Section H: References

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Section I: Appendices

Appendix A: Final Checklist

Item	Done?
Abstract (200-300 words, English & Arabic, no citations)	✓
Section A: Summary of previous work (within 2 pages)	✓
Section B: Methods and approaches (development phases)	✓
Section C: Activities (completed + not completed + why)	✓
Section D.1: Tools and technologies table with justification	✓
Section D.2: Interface type described and justified	✓
Section D.3: Screenshots with figure numbers and captions	✓
Section D.4: User manual for laymen (simple steps)	✓
Section D.5: Implementation progress table (70%+ Part I, 98%+ Final)	✓
Section E: Testing (plan, types, test cases, conclusion)	✓
Section F: Results and statistics	✓
Section G: Conclusion and future work	✓
Section H: References (APA + extra bibliography)	✓
Section I: Appendices	✓
File named: CPCS499_Group_AuditAI_Final	✓

Table 18: Final Report Checklist

Appendix B: System Architecture Overview

The AuditAI system follows a modular three-tier architecture:

- **Presentation Layer:** A web-based user interface that allows auditors to manage clients, upload financial documents, run audits, and review results through dashboards and visual reports.
- **Business Logic Layer:** A Python-based audit engine responsible for processing financial data, performing reconciliation and validation checks, detecting discrepancies, matching invoices, and generating audit outputs. This layer includes the Sales Audit, Purchases Audit, AR/AP Reconciliation modules, Smart Column Detector, and PDF validation components.
- **Data Layer:** A flexible file-based data system that supports Excel and PDF financial documents. The system generates structured outputs for dashboards and downloadable audit reports. Supabase is used for authentication, user management, and backend data handling.

Appendix C: Audit Engine Module Summary

Module	Input	Checks Performed	Output
Sales Audit	Sales invoices Excel	VAT validation, invoice sequence gaps, duplicate detection, total consistency checks	436 invoices processed with 0 validation errors
Purchases Audit	Purchase orders Excel	Same validation checks as Sales Audit	Validation completed; purchase PDFs not available
AR Reconciliation	AR opening/closing balance, sales, receipts	Reconciliation of opening balance, sales, receipts, and closing balance	37 customers analyzed; 6 exceptions detected
AP Reconciliation	AP opening/closing balance, purchases, payments	Reconciliation of opening balance, purchases, payments, and closing balance	12 suppliers analyzed; 8 exceptions detected
PDF Validation	Sales invoice PDFs + Excel data	PDF extraction, OCR fallback, and invoice field matching	438 invoices processed with 96.6% match rate

Table 19: Audit Engine Module Summary